

Securities Education Certificate



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Lecture 6

Fixed Income

Lecturers

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Fixed Income

Notes by **Don-Angelo Sfeir**

Lecturer, Securities Education Certificate 2022/23.

Introduction

Fixed income is the world's largest asset class. With an estimated value of over \$200T globally, it is almost three times the size of combined global equity market valuation. Fixed income refers to any type of investment under which the borrower or issuer is obliged to make payments of a fixed amount on a fixed schedule. For example, the borrower may have to pay interest at a fixed rate once a year and repay the principal amount on maturity.

The foundational fixed income security is the **bond**. There are many different types of bonds with various structures and uses. Fixed income is much less risky than equity investments and generates a steady source of income, depending on the terms bond holders may also have complete downside protection.

What are bonds?

Bonds essentially represent a loan that has been turned into a security which can be traded. A government or corporation borrows money from investors and issues bonds in return. The bonds represent a commitment by the issuer to repay the principal (amount back) at maturity (expiry) as well as interest payments at fixed intervals.

Bonds fall under two main categories: *Sovereign* and *Corporate*.

Sovereign Bonds

A sovereign bond is a debt security issued by a national government to raise money for financing government programs, paying down old debt, paying interest on current debt, and any other government spending needs. Sovereign bonds can be denominated in a foreign currency or the government's domestic currency.

Corporate Bonds

A corporate bond is a bond issued by a corporation in order to raise debt. Companies use the proceeds from bond sales for a wide variety of purposes, including buying new equipment, investing in research and development, buying back their own stock, paying shareholder dividends, refinancing debt, and financing mergers and acquisitions.

Bond characteristics

Due to the nature of bonds and their issuance, each bond will be different from the next in terms of legislation and terms & conditions. However, all bonds have the following characteristics (among others) that help investors and institutions understand the risk and return potential of a given bond.

Maturity

This is the date when the principal of the bond is paid back to bond holders and the company's bond obligation ends. Therefore, it defines the lifetime of the bond. A bond's maturity is one of the primary considerations an investor weighs against their investment goals and horizon.

Maturity is often classified in three ways:

- Short-term: Bonds that fall into this category tend to mature within one to three years.
- Medium-term: Maturity dates for these types of bonds are normally over ten years.
- Long-term: These bonds generally mature over longer periods of time.

Secured/Unsecured

A bond can be secured or unsecured. A secured bond pledges specific assets to bondholders if the company cannot repay the obligation. This asset is also called collateral on the loan. So if the bond issuer defaults, the asset is then transferred to the investor. A mortgage-backed security (MBS) is one type of secured bond backed by titles to the homes of the borrowers.

Unsecured bonds, on the other hand, are not backed by any collateral. That means the interest and principal are only guaranteed by the issuing company. These bonds return little of your investment if the company fails. As such, they are much riskier than secured bonds.

Liquidation Preference

When a firm goes bankrupt, it repays investors in order of the *Capital stack* as it liquidates. After a firm sells off all its assets, it begins to pay out its investors. Senior debt is debt that must be paid first, followed by junior (subordinated) debt. Equity holders get whatever is left, this includes investors holding shares.

Bonds are typically a senior debt hence their low relative risk compared to other derivatives.

Coupon Rate

The coupon amount represents interest paid to bondholders, normally annually or semiannually.

To calculate the coupon rate, divide the annual payments by the bond principle:

$$\text{Coupon Rate} = \frac{\text{Annual Coupon}}{\text{Par Value of Bond}}$$

Tax Status

While the majority of corporate bonds are taxable investments, some government and municipal bonds are tax-exempt, so income and capital gains are not subject to taxation. Tax-exempt bonds normally have a lower coupon rate than equivalent taxable bonds. An investor must calculate the tax-equivalent yield to compare the return with that of taxable counterparts.

Callability

Some bonds can be paid off by an issuer before maturity. If a bond has a call provision, it may be paid off at earlier dates, at the option of the company, usually at a slight premium to the principle. A company may choose to call its bonds if interest rates allow them to borrow at a better rate. Callable bonds also appeal to investors as they offer better coupon rates.

Risks of Bonds

Bonds are a great way to earn income because they tend to be relatively safe investments. But, just like any other investment, they do come with certain risks. Here are some of the most common risks with these investments.

Prepayment Risk

Prepayment risk is the risk that a given bond issue will be paid off earlier than expected, normally through a call provision. This can be bad news for investors because the company only has an incentive to repay the obligation early when interest rates have declined substantially. Instead of continuing to hold a high-interest investment, investors are left to reinvest funds in a lower interest rate environment.

Interest Rate Risk

Interest rates share an inverse relationship with bonds, so when rates rise, bonds tend to fall and vice versa. Interest rate risk comes when rates change significantly from what the investor expected. If interest rates decline significantly, the investor faces the possibility of prepayment. If interest rates rise, the investor will be stuck with an instrument yielding below market rates.

The greater the time to maturity, the greater the interest rate risk an investor bears, because it is harder to predict market developments farther out into the future.

Credit/Default Risk

Credit or default risk is the risk that interest and principal payments due will not be made as required. When an investor buys a bond, they expect that the issuer will make good on the interest and principal payments—just like any other creditor.

When an investor looks into corporate bonds, they should weigh out the possibility that the company may default on the debt, this includes looking at the bond rating. Safety usually means the company has greater operating income and cash flow compared to its debt. If the inverse is true and the debt outweighs available cash, the investor may want to stay away.

Bond Ratings

Most bonds come with a rating that outlines their quality of credit. That is, how strong the bond is and its ability to pay its principal and interest. Ratings are published and are used by investors and professionals to judge their worthiness.

Agencies

The most commonly cited bond rating agencies are Standard & Poor's, Moody's Investors Service, and Fitch Ratings. They rate a company's ability to repay its obligations. Ratings range from AAA to Aaa for high-grade issues very likely to be repaid to D for issues that are currently in default.

Bonds rated Aaa to Baa or above are called investment grade. This means they are unlikely to default and tend to remain stable investments. Bonds rated BB to Ba or below are called junk bonds—default is more likely, and they are more speculative and subject to price volatility.

Firms may not have their bonds rated, in which case it is solely up to the investor to judge a firm's repayment ability. Because the rating systems differ for each agency and change from time to time, changes in some bond's rating can happen leading to a significant price change of the bonds being traded.

Grade	Description	Category
Aaa	Prime	<i>Investment-grade</i> Unlikely to default Tend to remain stable investments
Aa1	High grade	
Aa2		
Aa3		
A1	Upper medium grade	
A2		
A3		
Baa1	Lower medium grade	
Baa2		
Baa3		
Ba1	Non-investment grade speculative	
Ba2		
Ba3		
B1	Highly speculative	
B2		
B3		
Caa1	Substantial risks	
Caa2	Extremely speculative	
Caa3	Default imminent with little prospect for recovery	
Ca		

C	In default	
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Bond Yields

Bond yields are all measures of return. Yield to maturity (YTM) is the measurement most often used, but it is important to understand several other yield measurements that are used in certain situations.

Yield to Maturity (YTM)

Yield to maturity (YTM) is the most commonly cited yield measurement. It measures what the return on a bond is if it is held to maturity and all coupons are reinvested at the YTM rate. Because it is unlikely that coupons will be reinvested at the same rate, an investor's actual return will differ slightly. Calculating YTM by hand is a lengthy procedure, so it is best to use Excel's RATE or YIELDMAT functions.

$$YTM = \sqrt[n]{\frac{\text{Face Value}}{\text{Current Value}}} - 1$$

Current Yield

The current yield can be used to compare the interest income provided by a bond to the dividend income provided by a stock. This is calculated by dividing the bond's coupon by the bond's current price.

Keep in mind, this yield incorporates only the income portion of the return, ignoring possible capital gains or losses associated with the principle. As such, this yield is most useful for investors concerned with current income only.

$$\text{Current Yield} = \frac{\text{Annual Coupon income}}{\text{Current Price}}$$

Nominal Yield

The nominal yield on a bond is simply the percentage of interest to be paid on the bond periodically. It is calculated by dividing the annual coupon payment by the principal value of the bond.

It is important to note that the nominal yield does not estimate return accurately unless the current bond price is the same as its principal value. Therefore, nominal yield is used only for calculating other measures of return.

$$\text{Nominal Yield} = \frac{\text{Annual Coupon income}}{\text{Principal}}$$

Yield to Call (YTC)

A callable bond always bears some probability of being called before the maturity date. Investors will realise a slightly higher yield if the called bonds are paid off at a premium. An investor in such a bond may wish to know what yield will be realised if the bond is called at a particular call date, to determine whether the prepayment risk is worthwhile.

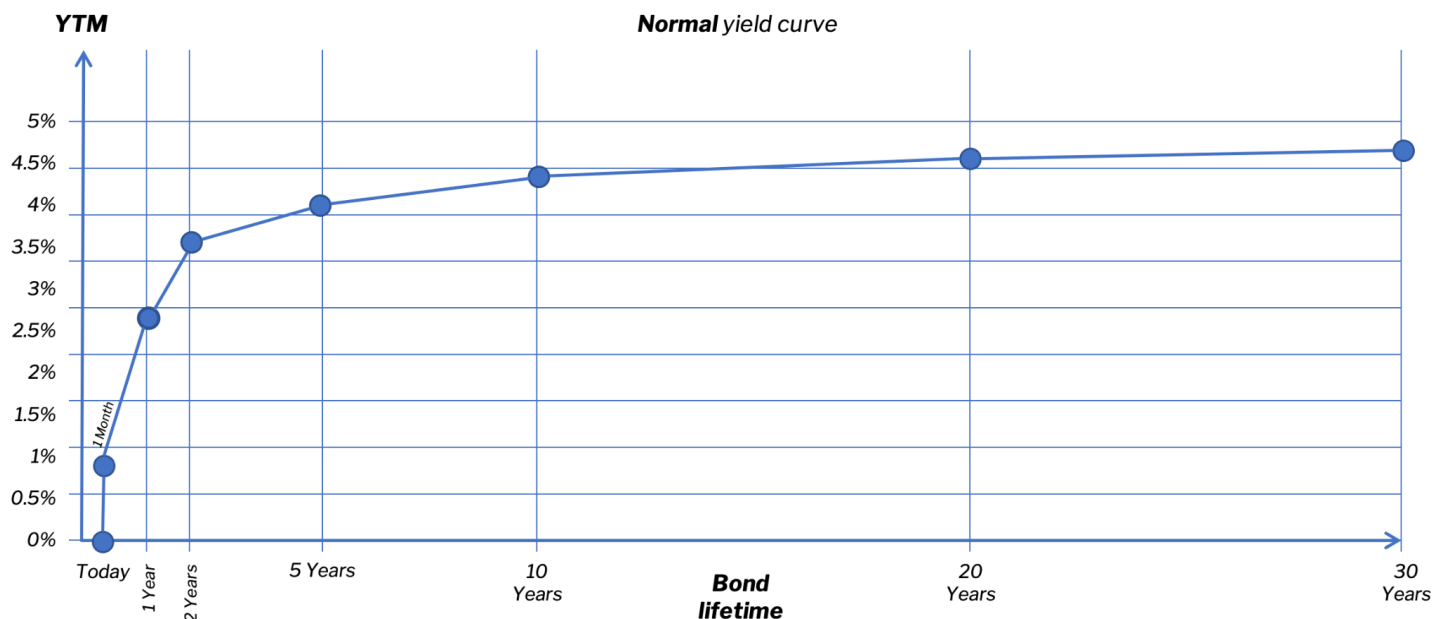
(Annual) Realised Yield (ARY)

The realised yield of a bond should be calculated if an investor plans to hold a bond only for a certain period of time, rather than to maturity. In this case, the investor will sell the bond, and this projected future bond price must be estimated for the calculation. Because future prices are hard to predict, this yield measurement is only an estimation of return.

Bond Spreads

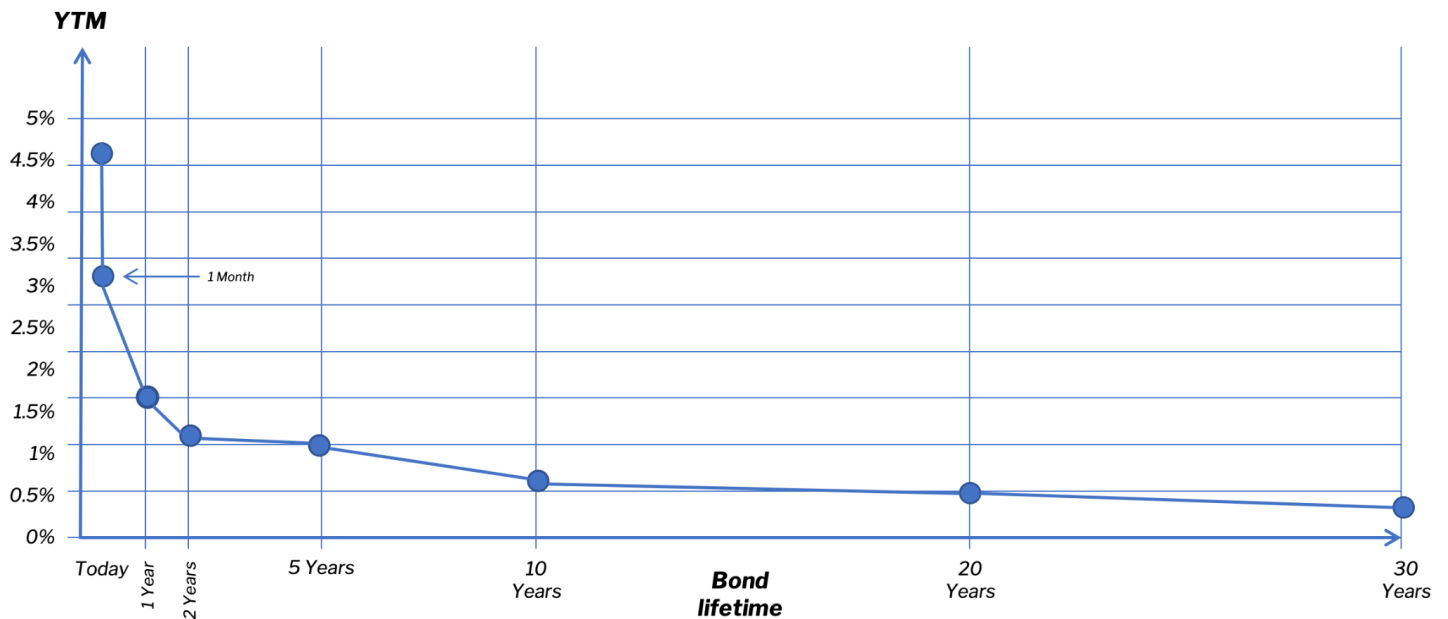
The yield curve

For a bond that has a lifetime of 1 year after issue, the yield is quite low. This is because the bond doesn't last very long so the issuer has little incentive to compensate the bondholder for a year's worth of debt. For successively longer bond lifetimes, the yields *tend* to increase. We can plot *hypothetical* bond yields for the UK Gilt as an example of this:



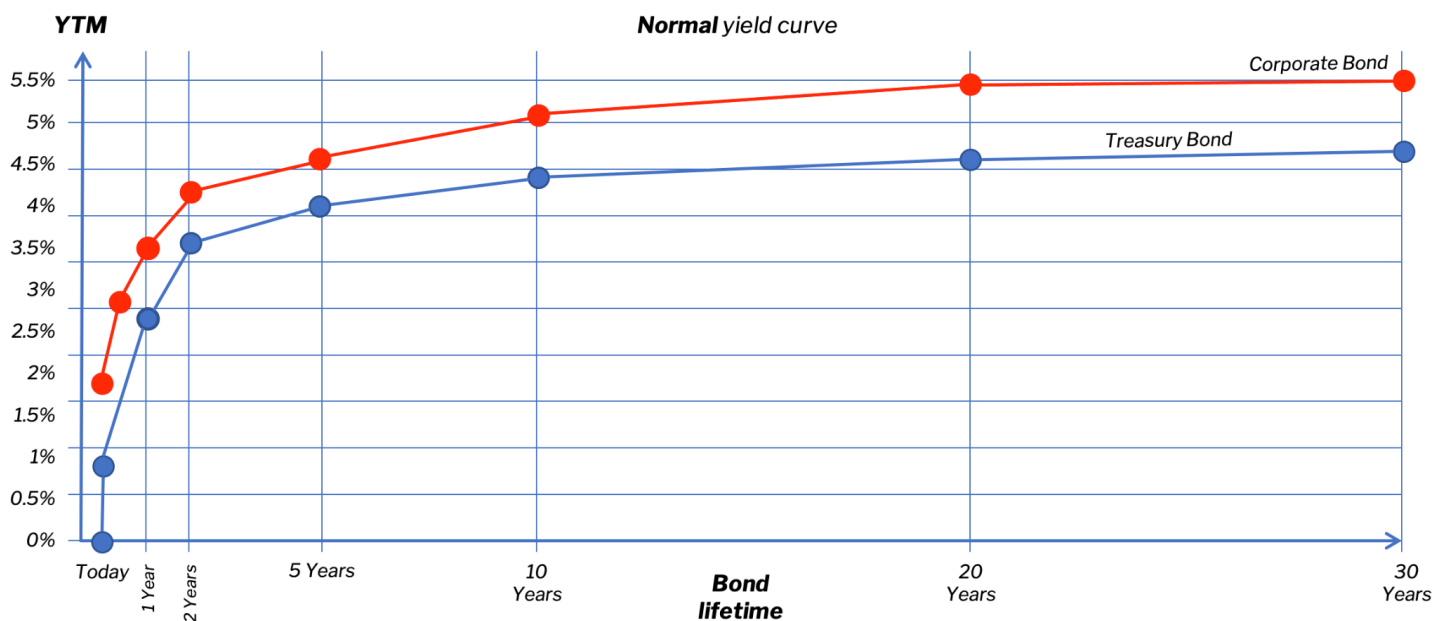
These types of yield curves are known as **normal** yield curves. You would expect most yield curves to curve upwards since the longer you hold a bond, the greater the return and hence the greater the YTM.

Sometimes, this might not be the case. Instead, the yield curve could flip, giving instead an **inverted** yield curve:



Inverted yield curves indicate recessions. Investors want to be guaranteed a safe return on their investment in bonds so they'll buy long term bonds if they believe a recession is coming. Even if a recession does happen, the central banks might have to lower interest rates and that would make the current long-term bonds even more desirable due to their higher coupon rate. To compensate for this, the central bank would have to negotiate higher yields in a bid to ensure people buy up their bonds and the government has enough money on its hands to deal with an oncoming recession, hence the *inverted* yield curve.

Because Governments can easily pay off any bonds it issues owing to its taxation system, corporations have to entice investors with yields much higher than those that the Government is offering. As a result, corporation bonds exist on a yield curve much higher than treasury bonds. For this reason, we also name treasury bonds 'benchmark yield curves':



Yield spreads

Corporate bonds versus treasury bonds can cause a few questions including the way we compare them. Investors tend to use two metrics to solve this problem: **G-spreads** and **Z-Spreads**. Let's discuss these in turn so we know what they mean.

G-spread

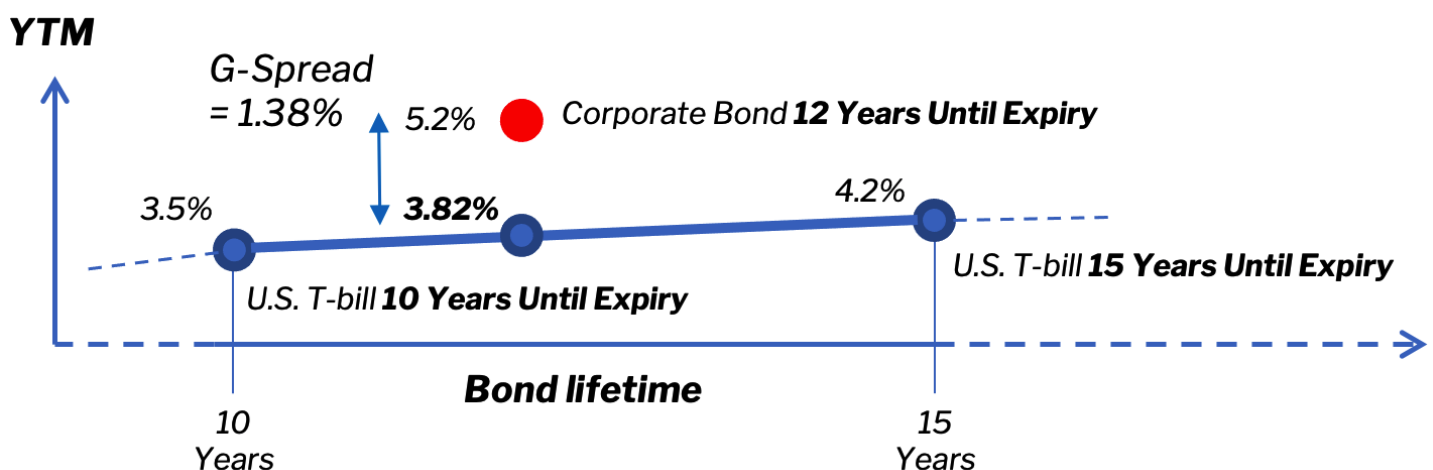
G-Spreads are the simpler ones to explain. Essentially, you are measuring the difference in yields between the corporate bond you are looking at and a corresponding treasury bond for a given duration before expiry. The G in G-spread simply stands for Government spread. Often, this spread can be written as:

$$\text{G-spread} = Y_c - Y_g$$

where Y_c is the corporate bond yield and Y_g is the government (i.e. treasury) yield.

Sometimes, it is possible that the maturity of a corporate bond doesn't match any government bonds that you find. In this case, you would have to use linear interpolation to get around the problem in finding a value for Y_g (as you shall see in the following example below).

Take a corporate bond with a 12 year duration until maturity and two U.S. T-bills as treasury bonds:



To find the YTM of the U.S. T-bill at 12 years, we would need to do some linear interpolation:

$$3.5 + \frac{2}{5} \times 0.8 = 3.82\%$$

Which gives us a hypothetical YTM of the U.S. T-bill for a 12-year lifetime of 3.82% assuming that the YTM of T-bills can be modelled linearly between those two points.

This then helps to calculate the G-Spread of the corporate bond:

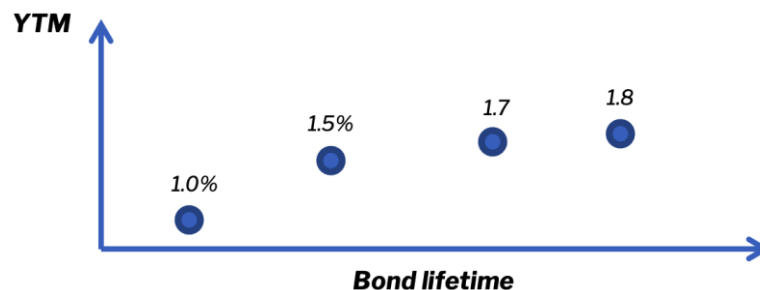
$$5.2 - 3.82 = 1.38\%$$

Z-spread

Remember calculating present values of equities in lecture 3? You can do the same with bonds too. The trouble is that the numbers we might use as discount rates will vary as the rate of return changes in line with

the interest rates offered by treasury bonds. To solve this, we might match the discount rates with the YTM of zero-coupon bonds (known as spot curves).

However, this can lead to another problem. Sometimes, the spot rates of e.g. T-bills may misprice a bond such that the present value of a bond is far higher than the market rate. Take the current spot rates of T-bills from the graph presented below:

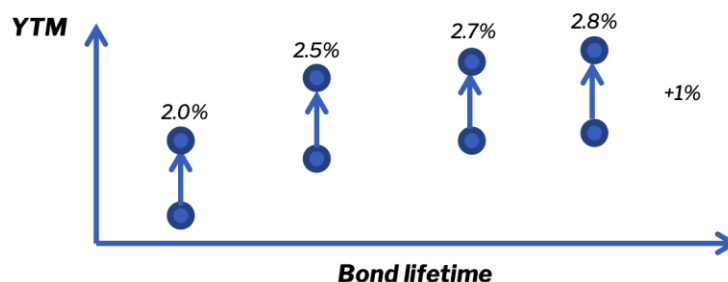


Say that there existed a 4-year 10% coupon rate £100 corporate bond which was trading on the market for £127. If we used the current spot rates above to calculate the present value for the bond, we'd get something completely different:

$$PV = \frac{10}{1 + 1\%} + \frac{10}{(1 + 1.5\%)^2} + \frac{10}{(1 + 1.7\%)^3} + \frac{10}{(1 + 1.8\%)^4} = £131.53$$

Clearly, the present value of the bond here is too high using the spot rates of the T-bills. If we, however, boosted these spot rates artificially so that the discount rate was higher, we might end up closer to the £127 that we're looking to match.

Let's try adding a constant 1% (i.e. 100 bps) to each of the spot rates:



You can see that there's a constant shift of 1% for all the spot rates.

Let's try again:

$$PV = \frac{10}{1 + 2\%} + \frac{10}{(1 + 2.5\%)^2} + \frac{10}{(1 + 2.7\%)^3} + \frac{10}{(1 + 2.8\%)^4} = £127.05$$

Now, we've got a much better discount rate that tells us the market expects a rate of return of 1% more than each of the treasury spot rates!

Looking at this more generally, we can write:

$$PV = \frac{10}{1 + r + Z} + \frac{10}{(1 + r + Z)^2} + \frac{10}{(1 + r + Z)^3} + \frac{10}{(1 + r + Z)^4}$$

where r is the spot rate and Z is the constant we added to the spot-rate. **This is the Z-spread.**

Professionally, it is referred to as *the spread that must be added to each spot interest rate to cause the present value of the bond cash flows to equal the bond's price.*

Now you can see why it makes sense.

N.B. You won't be expected to calculate Z-spreads in the exam at all!

It cannot be solved algebraically but instead through numerical methods i.e. trial and error.

Why Bonds over Equities

In general, investing in debt is relatively safer than investing in equity, and in turn the returns differ considerably. That's because debtholders have priority over shareholders in the capital stack, that is if a company goes bankrupt most if not all funds accrued through the liquidation of assets will go to repaying debt holders, including bond holders, leaving little not no reimbursement for the equity partners, share holders for example.

In terms of risk, bonds from developed governments, Treasury bonds, are considered risk-free as the entire economy of the country who issued the bonds would have to collapse and the nation declare bankruptcy for the bonds to default. While not exactly yielding high returns, in 2020, a 30-year US treasury bond yielded an interest rate of about 1.7%, there are cases where bonds yield much higher rates and are a more attractive investment. For example the current yield of the same 30 year US treasury bond is about 3.79%, over double that of 2020. This is because of the inflation crisis we are facing and how that translates to rising interest rates.

When a government wants to bring down or control inflation they raise interest rates, due to Covid, the energy crisis, and the Russo-Ukrainian war 2022 has seen unprecedented soars in inflation across the globe with most countries being hit regardless of their economic size or position. In an attempt to control this inflation governments, most notably the US, UK, EU, and Brazil, have hiked their interest rates by massive amounts almost monthly leading to some of the highest interest rates ever.

Bond prices are inversely proportional to interest rates, when interest rates rise bond prices drop. That is why we are seeing such unusually high yields in the bond market, as discussed in previous sections current yield would rise if bond prices drop. Aside from making bond prices low and yields high, rising interest rates raise the cost of capital so other investment tools become less attractive due to rising costs and higher risks.

Occasionally, yields can spike to very high levels due to the underlying bond price crashing. This happens at times of uncertainty and instability in the market. Most recently the UK's 10-year Gilts saw yields spike tremendously due to the crashing of the pound in late September and the new government's controversial tax cut plans.

More predictable returns

Stocks will outperform bonds in the long run. However, bonds outperform stocks at certain times in the economic cycle. It's not unusual for stocks to lose 10% or more in a year, so when bonds make up a portion of a portfolio, they can help smooth out the bumps when a recession comes along.

Also, in certain life situations, people may need security and predictability. Retirees, for instance, often rely on the predictable income generated by bonds. If your portfolio consisted solely of stocks, it would be quite disappointing to retire two years into a bear market. By owning bonds, retirees can predict with a greater

degree of certainty how much income they'll have in their later years. An investor who still has many years until retirement has plenty of time to make up for any losses from periods of decline in equities.

Better Than the Bank

The interest rates on bonds are typically greater than the deposit rates paid by banks on savings accounts. As a result, if you are saving and you don't need the money in the short term (in a year or less), bonds will give you a relatively better return without posing too much risk.

Bonds from stable governments and large corporations, Apple, Google, Amazon, are the best investment tools for capital preservation. However, keep in mind that although bonds are safer (as a rule) that doesn't mean they are all completely safe. There are also very risky bonds, which are known as junk bonds.

Asset Backed Securities

Asset-backed securities allow their issuers to raise cash. The underlying assets of an ABS are often illiquid and can't be sold on their own due to their risk. So, pooling assets together and creating a financial instrument out of them—a process called securitization—allows the issuer to make illiquid assets marketable to investors. The underlying cash flow obligations can come from many different industries and types, usually a security is made up of only one family of underlying assets.

Collateralized Loan Obligations (CLO)

A Collateralized Loan Obligation (CLO) is a type of security that allows investors to purchase an interest in a diversified portfolio of Corporate loans. The company selling the CLO will purchase a large number of corporate loans from borrowers such as private companies and private equity firms, and will then package those loans into a single CLO security.

Collateralized Mortgage Obligation

Collateralized mortgage obligations consist of several tranches, or groups of mortgages, organised by their risk profiles. The trenches typically have different principal balances, interest rates, maturity dates, and potential of repayment defaults. Collateralized mortgage obligations are sensitive to interest rate changes as well as to changes in economic conditions, such as foreclosure rates, refinance rates, and the rates at which properties are sold. Each tranche has a different maturity date and size and bonds with monthly coupons are issued against it funded by the mortgage repayments made on the underlying asset.

The 2008 Financial Crisis and CMOs

CMOs were complex and involved many different mortgage types. For many reasons, investors focused on the income streams offered by CMOs rather than the health of the underlying mortgages themselves. As a result, many investors purchased CMOs full of subprime mortgages, adjustable-rate mortgages, mortgages held by borrowers whose income wasn't verified during the application process, and other risky mortgages with high risks of default.

The use of CMOs has been criticised as a precipitating factor in the 2007-2008 financial crisis. Rising housing prices made mortgages look like fail-proof investments, enticing investors to buy CMOs and other Mortgage backed securities, but market and economic conditions led to a rise in foreclosures and payment risks that financial models did not predict. The aftermath of the global financial crisis resulted in increased regulations for mortgage-backed securities.

Collateralized Bond Obligation

A collateralized Bond Obligation (CBO) is an investment-grade bond that is backed by a pool of junk bonds. Because the pool includes several types of credit quality bonds together from multiple issuers, they offer enough diversification to be structured as "investment grade."

The role of Bonds in a Portfolio

Investors include bonds in their investment portfolios for a range of reasons some of which we have already briefly touched on...

Income generation

Bonds provide investors with a source of income in the form of coupon payments. The investor can use the income generated by their investments for spending or reinvestment. Shares also provide income in the form of dividends: however, such payments are less certain and tend to be less than bond coupons.

Capital preservation

Unlike stocks, the principal value of a bond is returned to the investor in full at maturity. This can make bonds attractive to risk-averse investors who are concerned about losing their capital.

Capital appreciation

Although bonds are often viewed as a capital preservation tool, they also offer opportunities for capital appreciation. This occurs when investors take advantage of rising bond prices by selling their holdings prior to maturity on the secondary market. This is often referred to as investing for total return and is one of the more popular bond investment strategies.

Hedge against economic slowdown

While investors in stocks typically do not welcome a slowdown in economic growth, it can be a good thing for bond investors. This is because slower growth usually leads to lower inflation, which makes bond income more attractive. An economic slowdown may also be negative for company profits and stock market returns, adding to the attractiveness of bond income during such a time.

Diversification

Bonds are considered a defensive asset class because they are typically less volatile than some other asset classes such as stocks. Many investors include bonds in their portfolio as a source of diversification to help reduce volatility and overall portfolio risk.